

Oliver Perrin

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SUMMARY

Founding machine learning engineer on a 5-person core team of a pre-launch startup, owning the ML system, production database, and backend API end-to-end; independently trained a compact domain-specialist model that outperforms frontier LLMs on the company's internal benchmark. Separately founded and launched LiminalML, a full-stack AI SaaS. Looking to keep building at the intersection of ML and product as an early technical leader.

EXPERIENCE

Machine Learning Engineer (Founding Team)

May 2026 – Present

Stealth Startup

Boone, NC

- Sole MLE on the 5-person core team of a pre-launch consumer product-comparison startup; retained through a 10 → 5 team consolidation and promoted from intern to full-time, owning all ML, data-infrastructure, and backend decisions ahead of launch.
- Trained and productionized a compact (~250M-parameter) domain-specialist model for product-data harmonization that outperforms GPT/Claude/Gemini baselines on a human-audited gold set of 50+ examples, at a fraction of their inference cost and latency; continuously improving quality through iterative pre-training, post-training, and architecture work.
- Built the evaluation and release infrastructure around the model: human-audited gold benchmarks, frontier-model baselines, regression gates governing checkpoint promotion, and end-to-end cloud GPU training runs (A100/H100).
- Designed and own the production data layer, a MongoDB Atlas database with a 38K+ harmonized product catalog, embedding vector store, and faceted search indexes, plus a recommendation and gap-fill system layered on it, serving the web app, iOS app, and browser extension.
- Built and deployed the company's backend product API from scratch (FastAPI, JWT auth, real-time harmonize-on-add, recommendation endpoints) and the model-serving service behind it, both live on cloud infrastructure; act as the integration point and technical advisor for the three client-facing engineering teams.

Founder

Mar 2026 – Present

LiminalML (liminalml.com)

Boone, NC

- Solo-built and launched (Product Hunt, July 2026) an AI-powered ML/SWE interview-prep SaaS: a streaming Claude-backed tutor teaches 157+ topics through a structured six-stage session with persisted, resumable state and inline math, code, and diagram rendering.
- Own the full stack and business end-to-end: Next.js/React/TypeScript frontend, Supabase (Postgres, auth, OAuth), Stripe subscription billing with free tier and trials, PostHog analytics, and per-session LLM unit-economics management.

Systems Engineering Intern

June 2023 – August 2024

Triaxis Power Consulting LLC

Charlotte, NC

- Scoped and deployed an IoT monitoring system (Raspberry Pi, thermocouples, potentiometers) tracking pipe temperature and thermal expansion in power plants for predictive maintenance.

PROJECTS

LexiMind: Multi-Task Learning for Literary & Academic Text | *PyTorch, Transformers*

Nov 2025 – Present

- Implemented a 272M-parameter encoder-decoder transformer from scratch (Pre-LN, RMSNorm, relative position bias, FlashAttention, KV-cache) and jointly trained it on summarization, topic classification, and multi-label emotion detection.
- Diagnosed negative transfer in naive multi-task training and designed two targeted interventions (learned attention pooling, temperature-based task sampling), improving emotion-detection F1 by 77% over the naive baseline and beating single-task and BERT-baseline specialists; full ablation study with gradient-conflict diagnostics and MLflow tracking.

EDUCATION

Appalachian State University

Boone, NC

B.S. Computer Science, Minor in Mathematics, Data Science Certificate

May 2026

- Relevant Coursework: Linear Algebra, Statistical Data Analysis, Numerical Methods, Advanced Reinforcement Learning, Applied Machine Learning, Theoretical Computer Science, Data Structures & Algorithms
- Senior Capstone (ML Lead): built the ML harmonization prototype adopted into production at the partnering startup which was the foundation of the full-time founding-team role above.

TECHNICAL SKILLS

Machine Learning: PyTorch, Transformers (Hugging Face), model pre-training & post-training, evaluation & benchmarking, embeddings & retrieval, recommendation systems, multi-task & transfer learning, NLP, RL, LLM APIs (Claude, Gemini)

Systems & Backend: Python, FastAPI, MongoDB (Atlas), PostgreSQL, Supabase, Docker, cloud GPU training (A100/H100), model serving & deployment (Railway), CI/CD, MLflow, Weights & Biases

Product & Frontend: TypeScript, Next.js, React, Tailwind, Stripe, PostHog, C/C++, Java, SQL, L^AT_EX